

CourseHub - Technical Specification & Architecture Document

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1. Executive Summary & Architecture Overview

1.1 Executive Brief

CourseHub is a course management system providing API contracts for authentication, course administration, and student enrollment. The platform utilizes a JWT-based security model to manage role-specific access for students and instructors, centering on a data pattern of courses containing ordered modules and tracked student enrollments.

1.2 Maturity Assessment

The project is in a REFINEMENT state. While the API contracts and entity relationships are well-defined, there is a critical lack of high-level business goals and a missing definition of the project scope, which creates a risk of scope creep regarding payment and content delivery systems.

1.3 Technical Stack

- JWT (JSON Web Token)
- Bearer Token Authentication Scheme
- Relational Database (implied by entity relationships and foreign key constraints)
- Async Email Service

1.4 Architectural Constraints

- **Authentication**: JWT Bearer tokens required in Authorization header.
- **Course Deletion**: Forbidden if active enrollments exist (HTTP 409 Conflict / ACTIVE_ENROLLMENT_CONSTRAINT).
- **Enrollment Logic**: Duplicate enrollments are prohibited.
- **Progress Tracking**: Automatic completion timestamp trigger when progress equals 100%.
- **Course Visibility**: Enrollment restricted to published courses only.

1.5 Critical Dependencies

- JWT for session management and identity verification.
- Strict foreign key dependence: Enrollment requires existing User and Course entities.
- Relational constraint: Course deletion is gated by the existence of active Enrollment records.
- Async email service for student registration welcome flow.

2. Architecture Workflows & Visual Diagrams

(No diagrams provided in source data)

3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifier	Type	Description	Source Section
API-AUTH-JWT	Non-Functional	Authentication must be handled via JWT Bearer tokens in the Authorization header.	API Overview
REQ-AUTH-REG	Functional	Allow students to self-register with email, password, and display name, triggering an async welcome email.	POST <code>`/api/v1/auth/register`</code>
REQ-COURSE-CREATE	Functional	Allow instructors to create courses with associated modules.	POST <code>`/api/v1/courses`</code>
RULE-ACTIVE-ENROLL	Constraint	ActiveEnrollmentConstraint: A course cannot be deleted if it has active enrollments (409 Conflict).	DELETE <code>`/api/v1/courses/{course_id}`</code>
REQ-ENROLL-STUDENT	Functional	Allow students to enroll in published courses, ensuring no duplicate enrollments.	POST <code>`/api/v1/enrollments`</code>
REQ-ENROLL-AUTOCOMplete	Functional	Automatically set completed_at timestamp when progress reaches 100%.	PUT <code>`/api/v1/enrollments/{enrollment_id}`</code>

3.2 Security Rules

- ****Authentication****: All protected endpoints require a ``Authorization: Bearer `` header (API-AUTH-JWT).
- ****Role-Based Access****:
- Registration is restricted to the Student role (REQ-AUTH-REG).
- Course creation is restricted to the Instructor role (REQ-COURSE-CREATE).
- Enrollment is restricted to the Student role (REQ-ENROLL-STUDENT).

3.3 Data Models

Entity ID	Entity Name	Description
ENT-USER	User	User entity containing id, email, display_name, role (student/instructor), and timestamps.
ENT-COURSE	Course	Course entity containing title, description, price_cents, published status, and associated modules.
ENT-MODULE	Module	Module entity linked to a course with a specific order.
ENT-ENROLLMENT	Enrollment	Enrollment entity linking a student to a course with progress tracking.

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Missing Section	Priority	Remediation Advice
Goals & Objectives	HIGH	The document defines the 'How' (API) but not the 'Why' (Business Goals). Add a section explaining the purpose of CourseHub.
Scope & Out-of-Scope	MEDIUM	Define what the API does NOT handle (e.g., payment processing, content delivery) to avoid scope creep.
Open Questions & Uncertainties	LOW	List any undecided API behaviors or future iterations.

4.2 Remediation & Workflow

The project must transition from the REFINEMENT state to a FINALIZED state by addressing the high-priority gaps in business objectives and scope definition. This will ensure that the technical implementation of the API aligns with the intended business value.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
ActiveEnrollmentConstraint	BUSINESS_DOMAIN	RULE-ACTIVE-ENROLL	A validation rule preventing the removal of a learning offering if linked student registrations exist, triggering a 409 status code.
CORS Standard	TECHNICAL_STACK	API-AUTH-JWT	The browser-level security mechanism governing cross-origin resource sharing for the provided API endpoints.
Cryptographic Hashing	TECHNICAL_STACK	REQ-AUTH-REG	The one-way mathematical transformation applied to user passwords during the registration process to ensure secure storage.
JWT	TECHNICAL_STACK	API-AUTH-JWT	A compact, URL-safe means of representing claims to be transferred between two parties, passed via the Bearer scheme in the Authorization header.